Digital Citizen Response System

Concept Brief
RIDC PT Innovation Hackathon 2026
Software Category
Focus Area: Smart Governance & Public Service Delivery

The Problem

Local ward and district government offices across Tanzania still register citizens and receive community service complaints — water, electricity, waste, roads, and security — through manual, paper-based, or word-of-mouth processes. Citizens have no reference number for their case, no visibility into its status, and often no confirmation a report was even received. Ward officers have no shared system to track what is outstanding in their own area, and district administrators have no real-time picture of service demand, resolution times, or recurring problem areas across wards.

Who It Affects

End-users: residents who need to register with local government and report community service issues — including citizens who rely on basic feature phones and have limited or no data access. **Decision-makers:** ward officers, who approve registrations and dispatch technicians to reported issues, and district/municipal administrators, who oversee service delivery across multiple wards and need visibility into performance and demand.

The Solution

DCRS is a web platform that gives ward offices a single system for citizen registration and a “One-Stop” services desk, backed by automatic SMS updates so no citizen is left wondering what is happening with their case. Citizens register once with their ID, contact, and location down to street/mtaa level; their ward officer reviews and approves them, with an SMS sent either way. Approved citizens can then file a service request in one of five categories (Water, Electricity, Waste, Roads, Security) from a simple portal. Ward staff assign a technician and visit time — again notifying the citizen automatically — and once resolved, the citizen is prompted to rate the service. Every action, on either side, is logged for full accountability, and officers and administrators get a live dashboard of registrations, open issues, and activity across their ward or district.

Technology Stack

- **Backend:** Python / Django.
- **Database:** PostgreSQL.
- **Frontend:** React with Tailwind CSS, built as a lightweight single-page application with code-splitting and a minimized production bundle to stay responsive on modest hardware and slower connections.
- **Communication:** integration with an SMS gateway API for outbound status updates and delivery confirmation.
- **Deployment:** Gunicorn and Nginx, served over SSL. Hosted on a secure, production-grade server setup, with all citizen data protected by encrypted (HTTPS) connections — the same standard used by banking and government websites.

What Success Looks Like

Success will be measured by: the share of citizen registrations reviewed within 24–48 hours; the SMS delivery confirmation rate for status updates; average time from an issue being reported to being resolved; average citizen satisfaction rating collected through the in-app feedback step; the number of wards actively using the system during a pilot period; and a measurable drop in repeat or unresolved complaints for the same household or location.

Low-Bandwidth & Low-End Device Readiness

DCRS's frontend is a React and Tailwind CSS single-page application built with a deliberately small footprint — code-splitting, lazy-loaded routes, and a minimized production bundle keep initial load light. The interface runs in a standard mobile browser without requiring an app install, and is designed to remain usable on 2G/3G connections and low-end Android devices. Because every meaningful update is also pushed by SMS, citizens without a smartphone or active data plan still receive the same status updates as those using the web portal, ensuring the service reaches its target population regardless of device or connectivity.